MIL-I-46175A(MR)
12 August 1983
SUPERSEDES
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2 February 1976

MILITARY SPECIFICATION

INSPECTION AND TESTING OF DUCTILE CAST IRON GRADE 100-70-03, ULTRASONIC

This specification is approved for use by the Army Materials and Mechanics Research Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains general and specific requirements and governs the procedure for 100% nondestructive inspection and attainment of improved mechanical properties of ductile cast iron through flaw size control and acceptable graphite form requirements.

1.2 Classification. This specification is intended for assuring component quality of as-cast and machined ductile iron castings. This specification does not apply to cast steels, welds or wrought material.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

VV-L-800 - Lubricating Oil, General Purpose, Preservative

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, US Army Materials and Mechanics Research Center, ATTN: DRXMR-SMS, Watertown, MA 02172 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Military

MIL-I-6868 - Inspection Process, Magnetic Particle

STANDARDS

Military

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-410 - Nondestructive Testing Personnel Qualification and Certification
MIL-STD-453 - Inspection, Radiographic

(Copies of specifications and standards required by manufacturers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM A247 - Plate I, Evaluating the Microstructure of Graphite in Iron Castings
ASTM A536 - Standard Specification for Ductile Iron Castings
ASTM E8 - Methods of Tension Testing of Metallic Materials
ASTM E10 - Standard Method of Test for Brinell Hardness of Metallic Materials
ASTM E142 - Standard Method for Controlling Quality of Radiographic Testing
ASTM E446 - Standard Reference Radiographs for Steel Castings Up to 2 Inches in Thickness

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Mechnical properties. Ductile iron castings inspected by this specification are intended to meet mechanical property requirements. Specimens which are representative of the casting shall conform to the minimum requirements as indicated in Tables I and II. Dimensions of "Y" blocks from which test bars are taken shall be in accordance with ASTM A 536. Location of samples for mechanical properties taken directly from the casting shall be mutually agreed upon by the procuring activity and the supplier.

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Table I

Mechanical properties of test bars, minimum

Tensile strength psi	Yield strength 0.2% offset, psi	Elonogation in 2 inches, percent
100,000	70,000	3

Table II

Mechanical properties of castings, minimum

Tensile strength psi	Yield strength 0.2% offset, psi	Elonogation in 2 inches, percent
90,000	65,000	2

3.1.1 Hardness. The hardness shall be within the range specified on the engineering part drawings. For Grade 100-70-03, the Brinell hardness number shall be in the range of 229 - 285.

3.2 Microstructure. The graphite component of the cast microstructure shall consist of a minimum 85% spheroidal graphite conforming to types I and II of ASTM A247. The balance may consist of type III and IV but may not contain types V, VI or VII or other microconstituents such as carbon flotation or massive carbides which would impair engineering properties or introduce machining difficulties. The carbide content shall not exceed 3% and should be uniformly distributed. For pictorial classification of graphite types refer to Plate I, graphite form type chart in accordance with ASTM A247.

3.3 Material surface condition. The machined surface upon which ultrasound velocity measurements are conducted shall have a surface finish from 85 rms to 125 rms.

3.4 Radiography. Radiography requirements for production castings shall be in accordance with the requirements of the applicable radiographic position drawing or the contract (see 6.2).

3.5 Magnetic particle. Magnetic particle requirements for production castings shall be in accordance with procedures specified on the drawing or the contract.

3.6 Ultrasonic velocity. Ultrasonic velocity requirements for production castings shall be in accordance with procedures as specified hereinafter. (See 4.4.5) The minimum ultrasound velocity shall be 214,500 in./sec.

3.7 Marking.

3.7.1 Identification. Castings shall be identified by a conspicuous color code as mutually agreed upon by the procuring activity and the supplier. Marking shall be applied by steel stamping on the ultrasonic terminal pad.

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3.8 Record of Inspection. All results of the inspection shall be appropriately recorded. All results shall be identified, filed and made available to the Government representative upon request.

3.9 Workmanship. The finished castings shall be free from hot tears, porosity, cold shuts, cracks and other injurious defects that may be detrimental to the strength or intended performance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Examinations.

4.2.1 Casting surface. Unless otherwise specified each casting shall be 100 percent visually inspected for surface defects and irregularities. All cracks, hot tears, cold shuts and porosity are cause for rejection. Casting surfaces shall be reasonably smooth and free of burned-on sand. Castings with major surface defects shall be evaluated for use prior to further processing.

4.2.2 Radiographic examination. Radiographic examination on production castings shall be performed in accordance with the requirements of the applicable drawings or contract.

4.2.2.1 Radiographic inspection lot. A lot for radiographic inspection shall consist of all castings of a specified design and alloy produced at one facility by the same production technique from one innoculated ladle.

4.2.2.2 Frequency of radiographic examination. A sampling inspection plan shall be performed only after satisfactory results have been obtained with a specific number of consecutive castings as mutually agreed upon by the procuring activity and supplier. After this requirement has been met, an average outgoing quality limit shall be performed as mutually agreed upon by the procuring activity and the supplier shall apply as shown in Table III.

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Table III

Sampling plan for continuous inspection
average outgoing quality limit (AOQL)

AOQL desired	Frequency of inspection	Initial 100% inspection
2%	Every 20th piece (5%)	76 consecutive pieces
1%	Every 20th piece (5%)	150 consecutive pieces
1%	Every 10th piece (10%)	115 consecutive pieces
1%	Every 5th piece (20%)	72 consecutive pieces
0.5%	Every 5th piece (20%)	145 consecutive pieces
0.5%	Every 10th piece (10%)	225 consecutive pieces

When any of the following occurrences arise, inspection shall recommence as specified in 4.2.2.2.

(a) Samples from a lot do not meet acceptance criteria.

(b) Any pattern change or major change of foundry technique.

(c) Start of a new foundry or furnace technique.

(d) A lapse of production in excess of 3 months.

4.2.2.3 Radiographic reference standards. Production radiographs for evaluation shall be compared to reference radiographs applicable to ASTM E446 for steel. Maximum aceptable defects of production radiographs shall not exceed the severity level for the applicable reference discontinuity type. The severity level or standard requirement for production castings shall conform to levels designated in the engineering drawing or in the contract.

4.2.3 Marking. Castings shall be examined for marking requirements. Non conforming castings shall be rejected.

4.2.4 Preparation for delivery. Castings shall be examined for compliance with preservation, packaging, packing and marking for shipment requirements. Castings which do not comply with Section 5 shall be rejected.

4.2.5 Workmanship. Castings shall be examined for workmanship. Castings which do not comply with 3.9 shall be rejected.

4.3 Tests.

4.3.1 Magnetic particle. Each casting shall be tested in accordance with MIL-I-6868. The procuring activity shall establish the acceptance criteria.

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4.3.2 Ultrasound velocity test. "This test is intended to provide a go, non-go type screening for all the variables affecting ultrasonic velocity in the subject cast iron, including percent nodularity."

4.3.2.1 Electronic equipment. The ultrasonic equipment shall be capable of producing 2.25 mhz and 5.0 mhz frequencies. For ease of operation and production-line inspection, an ultrasonic velocity system with digital display of velocity and computation circuits is preferred. Other systems which feature a cathode ray tube display may be used, however, an ability to read the video display to an accuracy of ± 1% of the full screen distance is required. The degree in accuracy of video trace displacements (time) measurement can be verified by connecting the ultrasonic-pulser output signal to an oscilloscope having a minimum band width of 10 megahertz.

4.3.2.2 Surface condition. Flat and parallel surfaces shall be utilized for the entry and opposite surface reflection of pulsed longitudinal sound waves. If flat and parallel surfaces are absent on the cast part, an adjoining flat pad shall be cast as an integral part of the casting. The attached pad shall be cast at a location which is subsequently machined. The dimensions of the cast inspection pad shall be established by mutual agreement between the procuring activity and the producer.

4.4 Test methods.

4.4.1 Mechanical properties. Specimens which are representative of the casting shall be tested in accordance with ASTM E8. Standard test specimens shall be machined from cast one inch "Y" blocks as specified in ASTM A536.

4.4.1.1 Hardness. Brinell hardness tests shall be conducted in accordance with ASTM E10 on each casting and the area designated by the drawing or as mutually agreed upon by the procuring activity and the supplier.

4.4.2 Microstructure. Specimens shall be prepared and examined as specified in ASTM A247.

4.4.3 Magnetic particle. The test should be conducted by the dry continuous method utilizing magnetizing techniques so as to reveal discontinuities having axes in any direction. As a minimum, the test magnetic fields should be applied in two subsequent perpendicular positions. Castings shall be demagnetized after inspection.

4.4.4 Radiographic. All radiographic procedures for production castings shall be in conformance with ASTM E142.

4.4.5 Ultrasound velocity.

4.4.5.1 Ultrasound velocity measurements shall be conducted using immersion or contact methods.

4.4.5.2 Sensitivity adjustment. The sensitivity control shall be adjusted such that the front surface trace is 80% to 95% saturated and the first back reflection is not less than 20 percent of full saturation.

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4.4.5.3 Reference standards. Equipment shall be calibrated using an acoustically stable material such as steel or ductile iron ultrasonic velocity masters before and after inspecting each lot of castings. If a steel master is used, the thickness shall be measured with a vernier micrometer to an accuracy of within 1%. The velocity of sound in steel (V_s) is equivalent to the ratio of the thickness (d) divided by the time (t).

$$V_s = d/t$$

With the search unit positioned over the steel sample immersed in water, adjust the video distance (seep controls on ultrasonic unit) to obtain a fixed interval between the first echo trace and first back reflection which is the apparent thickness. Record this displacement. Next, position the search unit over the ductile iron sample (The thickness of the steel sample should be identical to the ductile iron specimens.) or part, measure and record the actual and apparent thickness. The relative velocity (V_r) in ductile iron is equivalent to the ratio of the actual thickness (dc) to the apparent (da) thickness. Analyze the section thickness and velocity signals to obtain a single output value which is compared with a minimum value of 214,500 inches/second commensurate with 85% nodularity.

$$V_r = dc/da$$

$$V \text{ (ductile iron)} = (dc/da) \times 23400 \text{ in./sec.}$$

4.4.5.4 Frequency of inspection. The frequency of inspection for all parts or segments of a part as designated by the engineering drawing or procuring activity shall be 100 percent.

4.4.5.5 Nodularity acceptance. Acceptance of the part for nodularity shall be determined by the ultrasonic velocity test. The range of ultrasonic velocities for acceptable nodularity shall lie within 214,500 in./sec. to 225,000 in./sec.

4.5 General inspection provisions.

4.5.1 Certificate of compliance. When specified, the supplier shall furnish test bars or supply certification that the castings comply with specified requirements.

4.5.2 Personnel qualifications.

4.5.2.1 Laboratories and personnel performing radiographic inspection shall be qualified in accordance with MIL-STD-453.

4.5.2.2 Personnel performing magnetic particle inspection shall be qualified in accordance with MIL-STD-410D.

4.5.2.3 Personnel performing ultrasonic inspection shall be qualified in accordance with MIL-STD-410D.

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5. PACKAGING

5.1 Preservation. All machined surfaces shall be coated with a preservative oil conforming to VV-L-800 to ensure against corrosion during transportation. Machined surfaces shall be clean and free from moisture before coating is applied.

5.2 Packing. All castings shall be packed for shipment in such a manner as to ensure acceptance by common or other carrier, for safe transportation, at the lowest rate, to the point of delivery.

5.3 Marking for shipment. Unless otherwise specified, marking for shipment shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended Use.

6.2 Ordering data.

6.2.1 Acquisition documents. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Whether patterns are to be furnished by the casting manufacturer or the Government and provisions for final ownership of the patterns.
(c) Weight and dimensions of castings.
(d) Provisions for extra production castings for mechanical properties testing.
(e) When test bars are required.
(f) When certification is required.

6.3 Radiographic position drawing. A position drawing is a drawing that shows the film positions for those areas to be radiographed and should specify the applicable radiographic standards.

6.4 Grade 100-70-03 was previously listed as SAE Grade D7003.

Custodian:
Army - MR

Preparing activity:
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Review Activity:
Army - AL

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DOCUMENT IDENTIFIER AND TITLE
MIL-I-46175A Inspection and Testing of Ductile Cast Iron Grade 100-70-03, Ultrasonic

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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